Exact joint-measurability threshold for four coplanar qubit measurements

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Four projective spin measurements on a qubit become jointly measurable after sufficient common depolarization. For four distinct coplanar axes, a known construction gives a visibility threshold in terms of the successive angular gaps, but its optimality for unequal gaps has previously been supported only numerically. We prove that the construction is optimal throughout the strict ordered semicircle domain. The proof evaluates the unrestricted four-measurement Fourier criterion exactly. The vector part of the known parent measurement supplies a feasible four-support primal point, while an explicit affine dual function has matching value. Its only nontrivial feasibility conditions reduce to an elementary inequality for three ordered planar unit vectors. Consequently, if $0 = \alpha_1 < \alpha_2 < \alpha_3 < \alpha_4 < \pi$, the maximum common visibility is $[\sum_{j=2}^{4} \sin((\alpha_j - \alpha_{j-1})/2) + \cos(\alpha_4/2)]^{-1}$. The result assumes unbiased binary qubit effects and an unrestricted parent POVM; it makes no claim for unequal visibilities, biased effects, or noncoplanar measurements.

I. INTRODUCTION

Joint measurability asks whether the statistics of several measurements can be recovered as marginals of a single parent positive-operator-valued measure (POVM). For noisy qubit measurements this question gives a precise operational measure of how much common white noise is needed before one simultaneous readout can simulate all measurement choices [1–3].

For coplanar unbiased binary qubit measurements of a common visibility, Andrejic and Kunjwal constructed a parent POVM whose admissible visibility is set by a polygonal norm sum [2]. They conjectured that this sufficient condition is also necessary. Grinko and Uola subsequently gave a necessary-and-sufficient Fourier characterization for any finite family of unbiased binary qubit measurements [3]; however, using it still requires a nontrivial norm minimization. Most recently, Porto *et al.* found strong numerical evidence that the polygonal bound is tight for planar projective measurements [4].

Here we settle the four-measurement specialization analytically. We evaluate all eight norm channels and retain all four free triple-index Fourier vectors. The proof is a matching primal–dual certificate and therefore does not restrict a putative parent POVM to the support pattern of the known construction. The result fixes the exact common depolarizing visibility for four arbitrary distinct coplanar qubit axes. It does not address the arbitrary-number conjecture.

II. MODEL AND EXACT STATEMENT

Let $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$, with $\mathrm{tr}(\sigma_k \sigma_l) = 2\delta_{kl}$. Common rotations, independent outcome relabelings, and permutation of four distinct coplanar projective axes allow representatives

$$\boldsymbol{n}_j = (\cos\alpha_j, \sin\alpha_j, 0), \qquad 0 = \alpha_1 < \alpha_2 < \alpha_3 < \alpha_4 < \pi. \tag{1}$$

For outcome $a \in \{-1, +1\}$ and common visibility $\eta \in [0, 1]$, define

$$E^{(\eta)}_{a|j} = \frac{1}{2}\big(\mathbb{I}_2 + a\eta\, \boldsymbol{n}_j \cdot \boldsymbol{\sigma}\big). \tag{2}$$

Thus $1 - \eta$ is the depolarizing-noise fraction. The four POVMs are jointly measurable when there are sixteen positive semidefinite effects $G_{\boldsymbol{s}}$, $\boldsymbol{s} \in \{-1, +1\}^4$, such that

$$\begin{aligned} &\sum_{\boldsymbol{s}} G_{\boldsymbol{s}} = \mathbb{I}_2, \\ &\sum_{\boldsymbol{s}:s_j=a} G_{\boldsymbol{s}} = E^{(\eta)}_{a|j} \quad (j = 1, \ldots, 4;\ a = \pm 1). \end{aligned} \tag{3}$$

No rank, planarity, or support restriction is imposed on the parent; zero effects are allowed.

Introduce

$$D = \sum_{j=2}^{4} \sin\frac{\alpha_j - \alpha_{j-1}}{2} + \cos\frac{\alpha_4}{2}. \tag{4}$$

The main result is the following.

Theorem. *Under Eq.* (1)*, the effects in Eq.* (2) *admit an unrestricted parent POVM if and only if*

$$\eta D \le 1. \tag{5}$$

Consequently their maximum common visibility is

$$\eta_*(\alpha_2, \alpha_3, \alpha_4) = D^{-1}. \tag{6}$$

Figure 1 shows the two pieces of planar geometry used below. The centrally symmetric polygon is useful for interpreting D, whereas the ordered arc of four auxiliary unit vectors controls dual feasibility.

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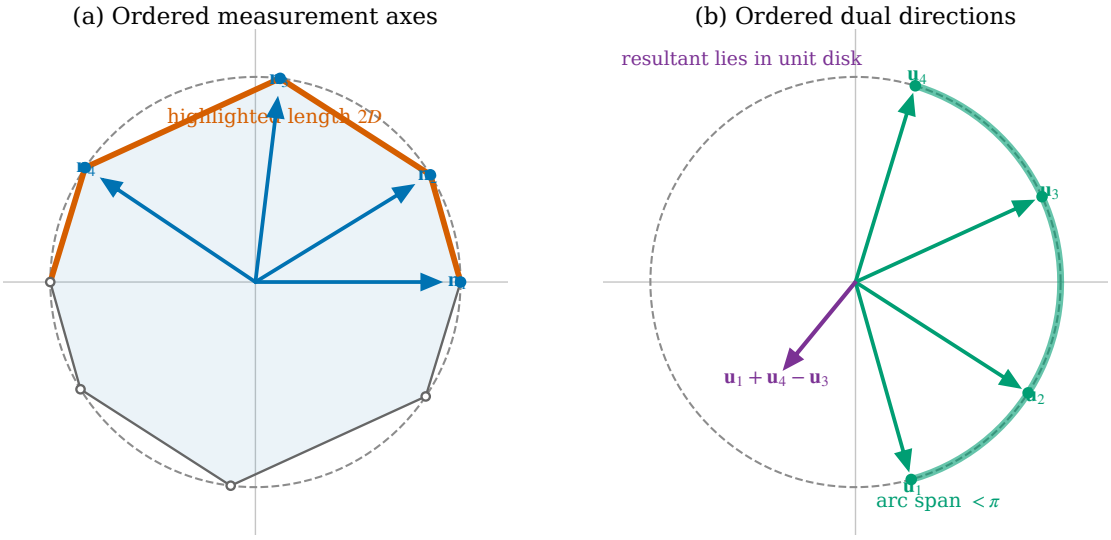


FIG. 1. Geometry of the exact certificate for a representative strict ordered configuration. (a) The highlighted path from $\boldsymbol{n}_1$ through $\boldsymbol{n}_4$ to $-\boldsymbol{n}_1$ has length $2D$; central symmetry therefore gives perimeter $4D$ for $\mathrm{conv}\{\pm\boldsymbol{n}_j\}$. (b) The four unit directions $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_4$ lie in angular order on an arc shorter than π. The displayed endpoints-minus-middle vector illustrates the lemma that bounds every off-support dual value by the unit disk. The figure is a schematic, not numerical evidence. It was produced with Matplotlib code developed with assistance from OpenAI Codex and checked against the analytic definitions.

III. FOURIER COMPATIBILITY CRITERION

We use the necessary-and-sufficient criterion of Grinko and Uola [3]. On the half cube

$$\mathcal{D}_4 = \{(1, \mu_2, \mu_3, \mu_4) : \mu_k \in \{-1, +1\}\}, \tag{7}$$

write

$$\boldsymbol{X} = (\boldsymbol{x}_{123}, \boldsymbol{x}_{124}, \boldsymbol{x}_{134}, \boldsymbol{x}_{234}) \in (\mathbb{R}^3)^4 \tag{8}$$

and define

$$q(\boldsymbol{n}_1, \ldots, \boldsymbol{n}_4) = \min_{\boldsymbol{X} \in (\mathbb{R}^3)^4} \sum_{\boldsymbol{\mu} \in \mathcal{D}_4} \|\boldsymbol{r}(\boldsymbol{\mu})\|, \tag{9}$$

$$\begin{aligned} \boldsymbol{r}(\boldsymbol{\mu}) = {} & \boldsymbol{n}_1 + \mu_2 \boldsymbol{n}_2 + \mu_3 \boldsymbol{n}_3 + \mu_4 \boldsymbol{n}_4 \\ & + \mu_2\mu_3 \boldsymbol{x}_{123} + \mu_2\mu_4 \boldsymbol{x}_{124} \\ & + \mu_3\mu_4 \boldsymbol{x}_{134} + \mu_2\mu_3\mu_4 \boldsymbol{x}_{234}. \end{aligned} \tag{10}$$

All four vectors in the minimization are unrestricted elements of $\mathbb{R}^3$. For the common-visibility effects, joint measurability is equivalent to the same minimum with fixed vectors $\eta \boldsymbol{n}_j$ being at most 8. Positive homogeneity of Eq. (9) gives

$$q(\eta \boldsymbol{n}_1, \ldots, \eta \boldsymbol{n}_4) = \eta\, q(\boldsymbol{n}_1, \ldots, \boldsymbol{n}_4) \tag{11}$$

for $\eta \geq 0$. It therefore remains to prove $q = 8D$ at unit visibility.

IV. MATCHING EXACT CERTIFICATES

Define the positive half-chord lengths and corresponding unit directions

$$d_j = \frac{\|\boldsymbol{n}_j - \boldsymbol{n}_{j+1}\|}{2}, \quad \boldsymbol{u}_j = \frac{\boldsymbol{n}_j - \boldsymbol{n}_{j+1}}{2d_j} \quad (j = 1, 2, 3), \tag{12}$$

$$c = \frac{\|\boldsymbol{n}_1 + \boldsymbol{n}_4\|}{2}, \qquad \boldsymbol{u}_4 = \frac{\boldsymbol{n}_1 + \boldsymbol{n}_4}{2c}. \tag{13}$$

Then $D = d_1 + d_2 + d_3 + c$.

A. Primal certificate

Define a vector-valued function R on $\mathcal{D}_4$ by the four nonzero values and set $R = 0$ at the other four vertices. To see directly that this is a feasible point of Eq. (9), denote the four displayed values by A, B, C, F in their

TABLE I. Support of the primal function R.

(μ_2, μ_3, μ_4)	$R(\boldsymbol{\mu})$
$(-,-,-)$	$8d_1\boldsymbol{u}_1 = 4(\boldsymbol{n}_1 - \boldsymbol{n}_2)$
$(+,-,-)$	$8d_2\boldsymbol{u}_2 = 4(\boldsymbol{n}_2 - \boldsymbol{n}_3)$
$(+,+,-)$	$8d_3\boldsymbol{u}_3 = 4(\boldsymbol{n}_3 - \boldsymbol{n}_4)$
$(+,+,+)$	$8c\boldsymbol{u}_4 = 4(\boldsymbol{n}_1 + \boldsymbol{n}_4)$

table order. Their normalized Walsh sums are

$$\begin{aligned}\tfrac{1}{8}(A+B+C+F)&=\boldsymbol{n}_1,\\ \tfrac{1}{8}(-A+B+C+F)&=\boldsymbol{n}_2,\\ \tfrac{1}{8}(-A-B+C+F)&=\boldsymbol{n}_3,\\ \tfrac{1}{8}(-A-B-C+F)&=\boldsymbol{n}_4.\end{aligned}\tag{14}$$

These are exactly the coefficients of the constant and the three singleton characters $1, \mu_2, \mu_3, \mu_4$. The remaining Walsh characters are $\mu_2\mu_3$, $\mu_2\mu_4$, $\mu_3\mu_4$, and $\mu_2\mu_3\mu_4$; their four coefficients can therefore be taken as the four free vectors in Eq. (10). Thus $\boldsymbol{r} = R$ is feasible and

$$q \le \sum_{\boldsymbol{\mu}} \|R(\boldsymbol{\mu})\| = 8(d_1+d_2+d_3+c) = 8D. \tag{15}$$

This feasible point is the Walsh transform of the vector part of the known parent construction [2]; it is not a support restriction on the minimization.

B. Planar unit-vector lemma

Let $\boldsymbol{u}(\theta) = (\cos\theta, \sin\theta, 0)$. If $a \le b \le c_0$ and $c_0 - a \le \pi$, set $p = b - a$ and $s = c_0 - b$. An elementary expansion gives

$$1 - \|\boldsymbol{u}(a) + \boldsymbol{u}(c_0) - \boldsymbol{u}(b)\|^2 = 8\cos\frac{p+s}{2}\sin\frac{p}{2}\sin\frac{s}{2} \ge 0. \tag{16}$$

Hence every endpoints-minus-middle combination of three unit vectors ordered on a closed semicircle has norm at most one.

C. Dual certificate

For every half-cube vertex define

$$\begin{aligned}\boldsymbol{Y}(\boldsymbol{\mu}) = &-\frac{\boldsymbol{u}_1+\boldsymbol{u}_4}{2} + \frac{\mu_2}{2}(\boldsymbol{u}_1-\boldsymbol{u}_2)\\ &+\frac{\mu_3}{2}(\boldsymbol{u}_2-\boldsymbol{u}_3) + \frac{\mu_4}{2}(\boldsymbol{u}_3-\boldsymbol{u}_4).\end{aligned}\tag{17}$$

Because $\boldsymbol{Y}$ contains only the constant and singleton characters, Boolean-character orthogonality gives

$$\sum_{\boldsymbol{\mu}\in\mathcal{D}_4} \chi_S(\boldsymbol{\mu})\boldsymbol{Y}(\boldsymbol{\mu}) = 0, \quad S \in \{123, 124, 134, 234\}, \tag{18}$$

where the four χ_S are the optimization characters appearing in Eq. (10).

The eight values of $\boldsymbol{Y}$ are listed in Table II. At the four support vertices of R they are $-\boldsymbol{u}_1, -\boldsymbol{u}_2, -\boldsymbol{u}_3, -\boldsymbol{u}_4$, respectively. To control the other four, note that the polar angles of the auxiliary directions are

$$\begin{aligned}\beta_1 &= \frac{\alpha_2}{2} - \frac{\pi}{2}, & \beta_2 &= \frac{\alpha_2+\alpha_3}{2} - \frac{\pi}{2},\\ \beta_3 &= \frac{\alpha_3+\alpha_4}{2} - \frac{\pi}{2}, & \beta_4 &= \frac{\alpha_4}{2}.\end{aligned}\tag{19}$$

TABLE II. The dual function. The first, fifth, seventh, and eighth rows are the support vertices of R in Table I.

(μ_2, μ_3, μ_4)	$\boldsymbol{Y}(\boldsymbol{\mu})$
$(-,-,-)$	$-\boldsymbol{u}_1$
$(-,-,+)$	$-(\boldsymbol{u}_1+\boldsymbol{u}_4-\boldsymbol{u}_3)$
$(-,+,-)$	$-(\boldsymbol{u}_1+\boldsymbol{u}_3-\boldsymbol{u}_2)$
$(-,+,+)$	$-(\boldsymbol{u}_1+\boldsymbol{u}_4-\boldsymbol{u}_2)$
$(+,-,-)$	$-\boldsymbol{u}_2$
$(+,-,+)$	$-(\boldsymbol{u}_2+\boldsymbol{u}_4-\boldsymbol{u}_3)$
$(+,+,-)$	$-\boldsymbol{u}_3$
$(+,+,+)$	$-\boldsymbol{u}_4$

They satisfy $\beta_1 < \beta_2 < \beta_3 < \beta_4$ and

$$\beta_4 - \beta_1 = \frac{\pi + \alpha_4 - \alpha_2}{2} < \pi. \tag{20}$$

Every off-support entry in Table II is therefore covered by Eq. (16). The support entries have norm one, so

$$\|\boldsymbol{Y}(\boldsymbol{\mu})\| \le 1 \qquad (\boldsymbol{\mu} \in \mathcal{D}_4). \tag{21}$$

For arbitrary values of all four free vectors in Eq. (10), Cauchy–Schwarz, Eqs. (18) and (21) imply

$$\begin{aligned}\sum_{\boldsymbol{\mu}} \|\boldsymbol{r}(\boldsymbol{\mu})\| &\ge -\sum_{\boldsymbol{\mu}} \boldsymbol{Y}(\boldsymbol{\mu})\cdot\boldsymbol{r}(\boldsymbol{\mu})\\ &= -\sum_{\boldsymbol{\mu}} \boldsymbol{Y}(\boldsymbol{\mu})\cdot(\boldsymbol{n}_1 + \mu_2\boldsymbol{n}_2 + \mu_3\boldsymbol{n}_3 + \mu_4\boldsymbol{n}_4).\end{aligned}\tag{22}$$

The fixed low-order coefficients in Eq. (14) allow the last line to be evaluated with R. At each nonzero vertex, $\boldsymbol{Y} = -\boldsymbol{u}_j$ and R is a positive multiple of $\boldsymbol{u}_j$, giving

$$-\sum_{\boldsymbol{\mu}} \boldsymbol{Y}(\boldsymbol{\mu})\cdot R(\boldsymbol{\mu}) = 8d_1 + 8d_2 + 8d_3 + 8c = 8D. \tag{23}$$

Equations (15) and (23) prove the global, unrestricted equality

$$\boxed{q(\boldsymbol{n}_1, \boldsymbol{n}_2, \boldsymbol{n}_3, \boldsymbol{n}_4) = 8D}. \tag{24}$$

This conclusion uses only weak duality; no numerical strong-duality assumption is involved. Combining Eqs. (11) and (24) with the Fourier compatibility criterion proves Eq. (5).

V. PHYSICAL RANGE AND CHECKS

The threshold lies in the physical visibility interval. Indeed, $2D$ is the total length of the four vectors

$$\boldsymbol{n}_1 - \boldsymbol{n}_2, \quad \boldsymbol{n}_2 - \boldsymbol{n}_3, \quad \boldsymbol{n}_3 - \boldsymbol{n}_4, \quad \boldsymbol{n}_4 + \boldsymbol{n}_1, \tag{25}$$

whose sum is $2\boldsymbol{n}_1$. The triangle inequality gives $D \geq 1$, and equality would require collinearity incompatible with the strict ordering in Eq. (1). Therefore $D > 1$ and $0 < \eta_* < 1$.

For the symmetric angles $(0, \pi/4, \pi/2, 3\pi/4)$, Eq. (4) gives $D = 4\sin(\pi/8)$ and recovers the known exact threshold $[4\sin(\pi/8)]^{-1}$ [1, 2]. The unequal-gap proof is exact and does not use a limiting expansion near this configuration.

Computations were used only to attack the certificate, not to establish the theorem. The accompanying verification program performs exact symbolic Walsh and trigonometric checks, solves the full four-vector eight-channel second-order-cone program, and separately optimizes sixteen unrestricted 2×2 Hermitian parent effects. Across 128 seeded hold-out angle sets, the maximum observed $|q_{\mathrm{SOCP}} - 8D|$ was 2.68×10^{-9} and the largest reported solver primal–dual gap was 5.72×10^{-10}. Across eight named direct-parent checks, the computed visibility agreed with D^{-1} within 7.9×10^{-9}. Residuals, tolerances, environment versions, and all sampled cases are recorded in the reproducibility package.

A. AI-assisted research and verification

OpenAI Codex, using GPT-6 Astra (Max reasoning) and GPT-5.6 Sol (Ultra reasoning) model configurations, was used substantively for literature organization, derivation discovery, code generation and debugging, symbolic and numerical checks, adversarial review, and manuscript drafting. The generated results were cross-checked within a staged workflow by independent source reconstruction, exact Walsh enumeration, symbolic factorization, and separately implemented numerical optimizations. These automated checks do not replace author judgment or accountability. This local draft requires the named author's personal scientific review, correction where needed, confirmation of the disclosure, and approval before any submission.

VI. SCOPE AND CONCLUSION

The exact formula applies to one qubit, four distinct coplanar projective axes represented in one open semicircle, binary unbiased effects, and one common depolarizing visibility. The parent POVM is unrestricted in $\mathbb{C}^2$ and may contain zero effects. Coincident or antipodal boundary points are outside the stated strict domain, although the formula has well-defined one-sided limits. Unequal visibilities, biased effects, noncoplanar directions, more or fewer than four measurements, higher-dimensional or nonbinary measurements, sequential nondisturbance, and multicopy notions require separate analyses.

Within this domain, the affine dual certificate supplies the missing necessity direction of the known parent construction. Thus the polygonal noise bound is not merely sufficient: for four axes it is the exact common white-noise threshold for simulation by a single parent measurement.

DATA AVAILABILITY

No experimental data were used. The source package accompanying this draft contains the verification code, frozen JSON output, plain-text log, pinned environment versions, and regeneration instructions. The analytic proof is self-contained and does not depend on the numerical calculations.

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OpenAI Codex also assisted substantively with manuscript organization and drafting. The research assistance and automated checks are described above; the named author's personal review, correction, and approval remain pending.

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